Jaehyeok Shim

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Education

Ulsan National Institute of Science & Technology (UNIST) — Ulsan, Republic of Korea
Ph.D. in Artificial Intelligence; Advisor: Prof. Kyungdon Joo — Mar. 2025 – Present

Ulsan National Institute of Science & Technology (UNIST) — Ulsan, Republic of Korea
M.S. in Artificial Intelligence; Advisor: Prof. Kyungdon Joo — Sep. 2021 – Aug. 2023
- Research contribution: Diffusion-Based Signed-Distance-Fields for 3D Shape Generation (CVPR 2023).

Seoul National University of Science and Technology — Seoul, Republic of Korea
B.S. in Electrical and Information Engineering; Double Major in Unmanned Vehicles Software — Mar. 2015 – Aug. 2021

Research Experience

Research Scientist Intern — Mar. 2025 – Jun. 2025
Adobe — United States
- Conducted research under the mentorship of Dr. Jaeshin Yoon.

Researcher — Sep. 2023 – Feb. 2025
Ulsan National Institute of Science & Technology (UNIST) — Ulsan, Republic of Korea
- Designed latent models for 3D compression and generation under Prof. Kyungdon Joo.
- Contributed to DITTO: Dual and Integrated Latent Topologies for Implicit 3D Reconstruction (CVPR 2024) and ContactGen: Contact-Guided Interactive 3D Human Generation for Partners (AAAI 2024).

Sergeant, Information Systems Specialist — Aug. 2016 – May 2018
Republic of Korea Army, 1st Division — Paju, Republic of Korea

Publications

DentiField: Implicit Surface Semantic Field for Intraoral Mesh Segmentation | WACV 2027
- **Jaehyeok Shim**, Soomin Ham, Kyungdon Joo.
- Direct intraoral mesh segmentation that repurposes a pretrained implicit 3D model (TripoSG) as a surface semantic field.

Trip-to-Gaussian: A Versatile Framework for Unconditional 3D Generation | TMLR 2026
- Youngwoo Jeon, Inhyeok Choi, **Jaehyeok Shim**, Sangjune Park, Kyungdon Joo. [Project] [Paper] [Code]
- Coarse-to-fine unconditional 3D generation that converts triplane latents into high-quality 3D Gaussians.

DITTO: Dual and Integrated Latent Topologies for Implicit 3D Reconstruction | CVPR 2024
- **Jaehyeok Shim**, Kyungdon Joo. [Project] [Paper] [Code]
- Point-cloud-to-mesh reconstruction using complementary point and grid latent representations.

ContactGen: Contact-Guided Interactive 3D Human Generation for Partners | AAAI 2024
- Dongjun Gu, **Jaehyeok Shim**, Jaehoon Jang, Changwoo Kang, Kyungdon Joo. [Project] [Paper] [Code]
- Guided-diffusion model that generates human poses with physically plausible partner interactions.

Diffusion-Based Signed-Distance-Fields for 3D Shape Generation | CVPR 2023
- **Jaehyeok Shim**, Changwoo Kang, Kyungdon Joo. [Project] [Paper] [Code]
- Coarse-to-fine diffusion framework for high-fidelity 3D shape generation from signed-distance fields.

Academic Contributions

Peer Reviewer — Mar. 2024 – Oct. 2026
- WACV 2027, AAAI 2027, NeurIPS 2026, ECCV 2026, CVPR 2026, NeurIPS 2025, ICCV 2025, CVPR 2025, IEEE TIP 2025

Open-Source Projects

mesh2sdf-triton | PyTorch, Triton

- GPU-only, differentiable mesh-to-SDF library derived from Mesh2SDF, with Triton kernels and vertex gradients; about 34–44× faster than the original C++ CPU implementation on watertight meshes.

kitsu | PyTorch, DDP

- PyTorch boilerplate stack with a distributed-data-parallel trainer similar to PyTorch Lightning.

space-filling-pytorch | Triton

- Library implementing space-filling curves, including Hilbert curves and Z-order, in Triton.

fast-GeM | Triton

- Triton implementation of generalized mean pooling.

GEGLU-triton | Triton

- Triton implementation of gated exponential linear units.